

22GE004–BASICS OF ELECTRONICS ENGINEERING

**UNIT II & SIGNAL CONDITIONING USING
DIODE**

Review of PN Junction Diode

- It is possible to manufacture a single piece of a semiconductor material, half of which is doped by P-type impurity and the other half by N-type impurity as shown in Figure.1

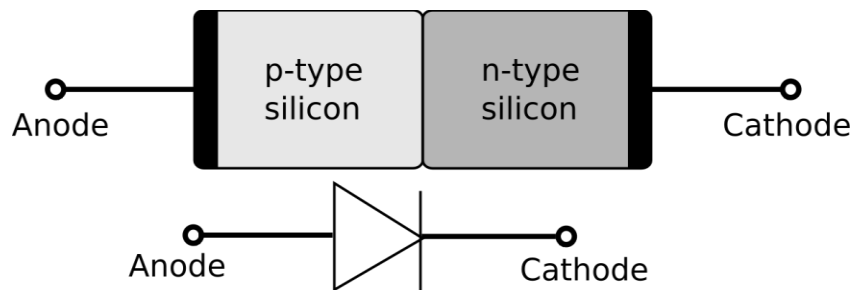


Figure1. Formation of PN Junction

- The plane dividing the two zones is called Junction.
- The P-N junction is fundamental to the operation of diodes, transistors and other solid-state devices.
- It is found that following three phenomena take place during the Junction formation
 - A thin depletion layer or region (also called space-charge region or transition region) is established on both sides of the junction and is so called because it is depleted of free charge carriers.
 - A barrier potential or junction potential is developed across the junction.
 - The presence of depletion layer gives rise to junction capacitance and diffusion capacitances.
- These phenomena are illustrated in the below figure

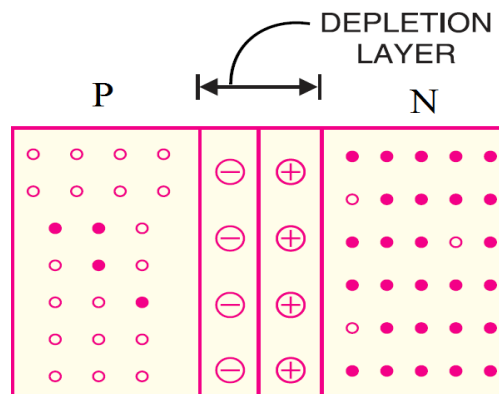


Figure2. Formation of Depletion Layer

1. Formation of Depletion Region in PN Junction

- Suppose that a junction has just been formed.
- At that instant, holes are still in the P-region and electrons in the N-region.
- However, there is greater concentration of holes in P-region than in N-region
- Similarly, concentration of electrons is greater in N-region than in P-region
- This concentration difference establishes density gradient across the junction resulting in carrier diffusion.
- Holes diffuse from P to N-region and electrons from N to P-region and this results in recombination at the junction.
- This recombination of free and mobile electrons and holes produces the narrow region at the junction called depletion layer.
- It is so named because this region is depleted of free and mobile charge carriers like electrons and holes—there being present only positive ions which are not free to move.
- The impurity atoms which provide the semigratory electrons and holes are left behind in an ionized state bearing a charge which is opposite to that of the departed carrier.
- These impurity ions, just like Germanium or Silicon atoms, are fixed in their positions in the crystal lattice in the P- and N- regions of the diode.
- This is shown in the below figure.

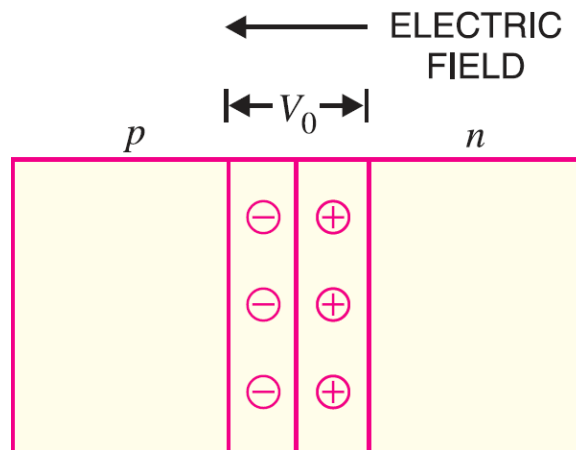


Figure 3. Depletion region and Barrier Potential

- They form parallel rows or 'plates' of opposite charges facing each other across the depletion layer.
- These fixed ions will restrict further movements of the electrons and holes through the depletion region.
- Ultimately, an equilibrium condition is reached when the depletion layer has widened to such an extent that no electrons or holes can cross the P-N junction.

- This shows the Depletion Region is formed in a PN junction.

2. Barrier Voltage or Junction Voltage in a PN Junction

- In the Depletion region, it can be seen that it has oppositely charged fixed rows of ions present in both P region and N region
- Because of this charge separation, an electric potential difference V_B is established across the junction
- It is known as junction or barrier potential.
- It stops further flow of carriers across the junction unless supplied by energy from an external source.
- At room temperature of 300°K, V_B is about 0.3V for Ge and 0.7V for Si.
- The value of Barrier Voltage is given by the equation

$$V_B = V_T \ln \frac{N_a N_d}{n_i^2} \quad (1)$$

- Where

V_B = Barrier Voltage

$V_T = \frac{kT}{e}$
 k = Boltzmann constant
 T = Temperature in °K
 e = charge of electron

N_a = Concentration of Acceptor atoms
 N_d = Concentration of Donor atoms

n_i = Intrinsic charge carrier concentration

- From the above equation, it can be seen that the Barrier voltage is directly proportional to the Temperature and doping concentration
- The formation of Barrier Potential can be summarised as follows
 - (a) As soon as P-N junction is formed, free electrons and holes start diffusing across the junction and recombining.
 - (b) Their recombination leads to the appearance of a depletion layer across the junction which contains no mobile carriers but only immobile ions.
 - (c) These immobile ions set up a barrier potential and hence an electric field which sets up drift current that is equal and opposite to the diffusion current when final equilibrium is reached.

3. Biasing of PN Junction Diode

- The process of connecting the P and N junction to an external supply to establish a current flow in the PN junction is known as biasing
- Based on the connection to the supply terminals, the biasing can be divided into two types
 - (a) Forward Biasing
 - (b) Reverse Biasing

3.1. Forward Biasing

- Suppose, positive battery terminal is connected to P-region of a semiconductor and the negative battery terminal to the N-region as shown in the below Figure

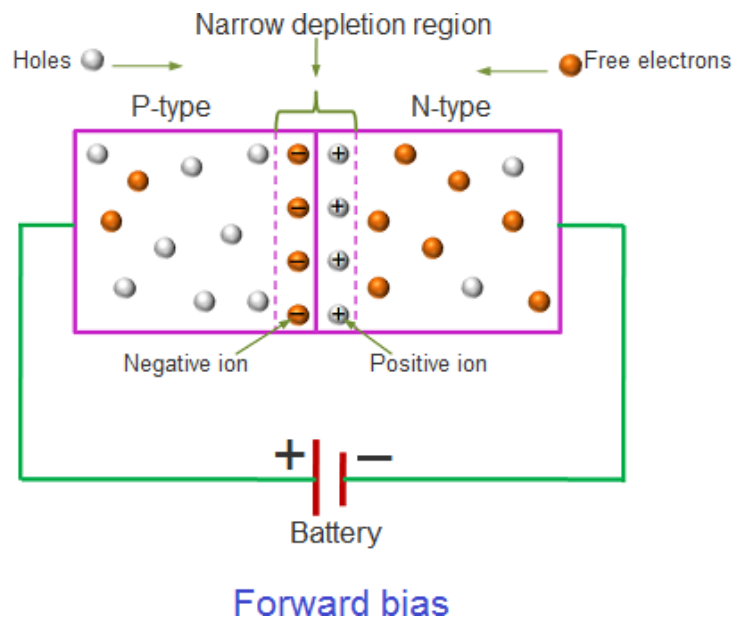


Figure 4. Forward Biasing of PN Junction

- In that case the junction is said to be biased in the forward direction because it permits easy flow of current across the junction.
- The forward bias of V volts lowers the barrier potential to $(V - V_B)$ which now allows more current to flow across the junction
- The mechanism of Forward Biasing can be explained as below:
 - As soon as battery connection is made, holes are repelled by the positive battery terminal and electrons are repelled by the negative battery terminal with the result that both the electrons and the holes are driven toward the junction where they

recombine.

- This movement of electrons to the left and that of holes to the right of the junction constitutes a large current flow through the semiconductor.
- The junction offers low resistance in the forward direction.
- It may be noted that though there is movement of both electrons and holes inside the crystal, only free electrons move in the external circuit i.e. in the battery-connecting wires.
- There is also present an extremely small amount of current I_0 due to minority carriers on either side of the junction.
- But it is negligible as compared to forward current which is due to majority carriers.
- This current I_0 depends on temperature but is almost independent of applied voltage.
- It may be noted that forward bias reduces the thickness of the depletion layer as shown in Figure 4.

3.2. Reverse Biasing

- When negative terminal of the battery is connected to N region and positive terminal of the battery is connected to the P region, the junction is said to be reverse-biased.
- In this case, holes are attracted by the negative battery terminal and electrons by the positive terminal also that both holes and electrons move away from the junction and away from each other.
- This is illustrated in the figure given below

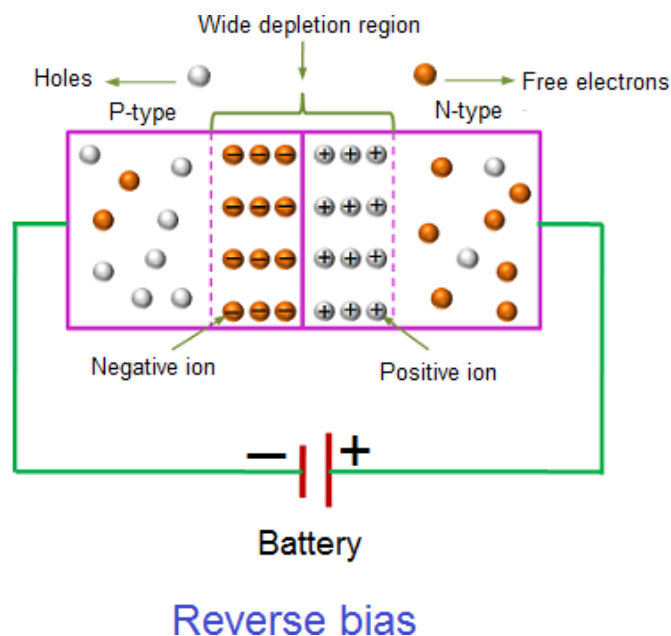


Figure 5. Reverse Biasing of PN Junction Diode

- Since there is no electron-hole combination, no current flows and the junction offers high resistance.
- In this case, the applied voltage increases the barrier potential to $(V + V_B)$, thereby blocking the flow of majority carriers.
- In this case, there is practically no current due to majority carriers, yet there is a small amount of current (a few μA only) due to the flow of minority carriers across the junction.
- Due to thermal energy, there are always generated some holes in the N-type region and some electrons in the P-type region of the semiconductor.
- The battery drives these minority carriers across the junction thereby producing a small current called reverse current or reverse saturation current I_0 or I_S .
- Since minority carriers are thermally-generated, I_0 is extremely temperature dependent.
- The name saturation has been used because we cannot get minority current more than what is produced by thermal energy.
- In other words, I_S does not increase with increase in reverse bias.
- I_S is found to increase approximately 7 percent per $0^\circ C$ rise in temperature both for Ge and Si.
- Reverse current doubles for every $10^\circ C$ rise in temperature.
- Reverse saturation current is also referred to as leakage current of the P-N junction diode.

Static Characteristics of Practical and Ideal Diode

1. Characteristics of Practical Diode

- Figure 1 given below depicts the Voltage Current Characteristics of a Low Power PN Junction Diode
- The characteristics have two parts
 - Forward Characteristics
 - Reverse Characteristics

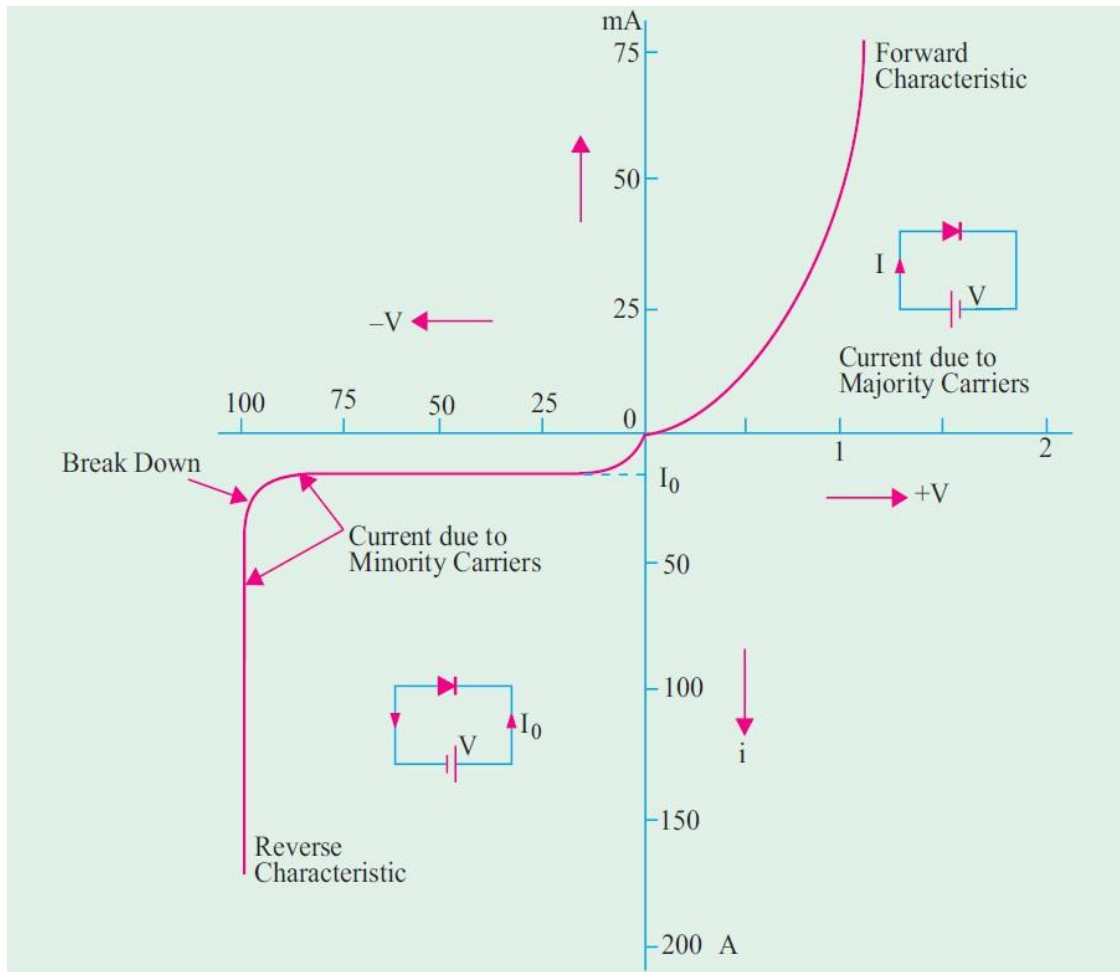


Figure 1. Voltage-Current Characteristics of Practical Diodes

1.1. Forward Characteristics

- When a Diode is forward biased and the applied voltage is increased from zero, no current will flow through the device in the beginning.
- This is because the external voltage is opposed by the internal barrier voltage V_B .
- The value of Barrier Voltage for Silicon is 0.7V.
- The value of Barrier Voltage for Germanium is 0.3V.
- As soon as the Barrier Voltage V_B is neutralized by the applied voltage, the current through the diode increases rapidly with the increasing applied voltage.
- From the figure, it is clear that even a very small voltage of 1V can produce a forward current of 50mA.
- The device will be destroyed if the applied voltage is increased beyond a certain

safelimit

- The movement of holes and electrons in a forward biased diode is shown in Figure 2.

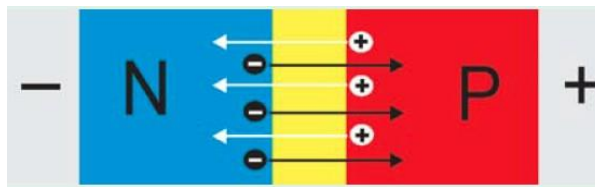


Figure2.ForwardBiasedPNJunctionDiode

1.2. ReverseCharacteristics

- When a Diode is reverse biased, the majority carriers are attracted towards the terminals and only a small current will flow through the diode due to the minority carriers
- When the voltage is increased in the reverse bias, the reverse current reaches its maximum very quickly
- The maximum current in the reverse bias is known as Reverse Saturation Current or Leakage Current I_0
- ItisintherangeofNanoamperesforSiliconandMicroamperesforGermanium
- Fromthecharacteristics,itisclearthatthevalueoftheReverseSaturationcurrent I_0 is independentoftheappliedreversevoltage
- However,theReverseSaturationCurrentdependson
 - (a) Temperature
 - (b) DegreeofDoping
 - (c) PhysicalsizeoftheJunction
- When reverse voltage exceeds a certain value called Reverse break-down voltage called V_{BR} or Zener voltage V_Z , the reverse current increases suddenly andsharply.
- Anyfurtherincreaseinvoltageislikelytodestroythedevice
- The movement of holes and electrons in a reverse biased diode is shown in Figure 3.

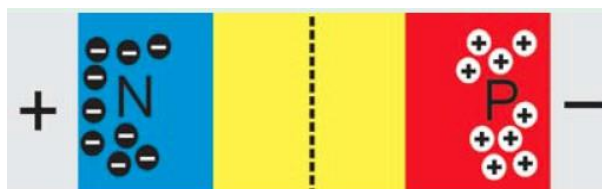


Figure3.ReverseBiasedPNJunctionDiode

1.3. MathematicalExpressionofStaticCharacteristics

- TheStaticCharacteristicsofapacticalPNjunctiondiodecanbeexpressedusing "BoltzmanDiodeEquation"
- TheBoltzmanDiodeEquationisgivenas

$$I = I_0 (e^{\frac{eV}{\eta kT}} - 1) \quad (1)$$

- Where
 - I_0 = Diode Reverse Saturation Current
 - V = Voltage across the junction
 - * Positive for Forward Biasing
 - * Negative for Reverse Biasing
 - k = Boltzmann constant ($1.38 \times 10^{-23} \text{ J/}^\circ\text{K}$)
 - T = Crystal Temperature in $^\circ\text{K}$
 - $\eta = 1$ for Germanium
 - $\eta = 2$ for Silicon
- Now, we can see that

$$\frac{e}{k} = 11,600 \quad (2)$$

- If we substitute equation (2) in equation (1)

$$I = I_0 (e^{\frac{11,600V}{\eta T}} - 1) \quad (3)$$

- If we substitute

$$\frac{T}{11,600} = V_T \quad (4)$$

- The equation will be modified as

$$I = I_0 (e^{\frac{V}{\eta V_T}} - 1) \quad (5)$$

- Consider the Room temperature in Kelvin ($273 + 20 = 293^\circ\text{K}$)
- Substituting this value in equation (4), we get

$$V_T = \frac{293}{11,600} = 25\text{mV} \quad (6)$$

- By substituting the value of V_T and η , the equation (5) can be written as below
- For Germanium

$$I = I_0 (e^{40V} - 1) \quad (7)$$

- When $V > 1$, the equation can be written as

$$I = I_0 e^{40V} \quad (8)$$

- For Silicon

$$I = I_0 (e^{20V} - 1) \quad (9)$$

- When $V > 1$, the equation can be written as

$$I = I_0 e^{20V} \quad (10)$$

2. Characteristics of Ideal Diode

- There is no such thing as an ideal diode or perfect diode.
- The existence of an ideal diode is visualized only for analysis purpose
- An ideal diode may be defined as a two-terminal device which
 - (a) Conducts with zero resistance when forward-biased
 - (b) Appears as an infinite resistance when reverse-biased
- Such a device acts as a short-circuit in the forward direction and as an open-circuit in the reverse direction as shown in Figure 4

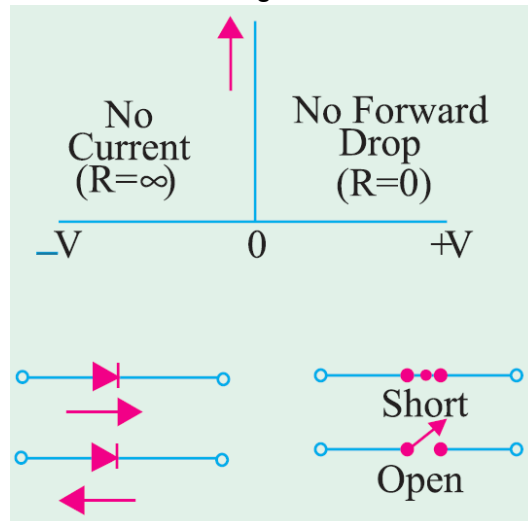


Figure 4. Characteristics and Equivalent of ideal diode

- There is no voltage drop across the diode as the diode offers zero resistance
- There is no reverse current because reverse resistance is infinite.
- Ideal diode can be considered as a bistable switch (ON and OFF) which is closed in the forward direction and open in the reverse direction

Junction Resistance and Junction Capacitance

1. Junction Resistance

- The depletion region present in a diode acts like barrier to electric current. Hence, it offers resistance to the electric current.
- The two types of resistance take place in the p-n junction diode are
 - Forward Resistance
 - Reverse Resistance
- The mathematical expression for the Junction Resistance is given as

$$r_j = \frac{dV}{dI} \quad (1)$$

- Also the conductance is given as

$$g_j = \frac{dI}{dV} \quad (2)$$

- Now the Diode current is given as

$$I = I_0 \left(e^{\frac{V}{V_T}} - 1 \right) \approx I_0 e^{\frac{V}{V_T}} \quad (3)$$

- Substituting equation (3) in equation (2)

$$g_j = \frac{\left(dI_0 e^{\frac{V}{V_T}} \right)}{dV} = \frac{I_0 e^{\frac{V}{V_T}}}{V_T} = \frac{I + I_0}{V_T} \quad (4)$$

- Figure 1. given below represents the various Resistances present in a PN junction diode

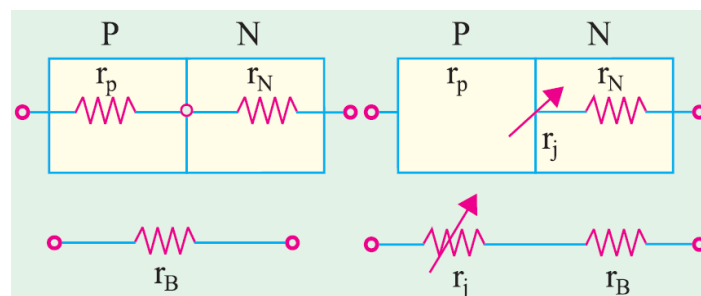


Figure 1. Resistances in a PN Junction Diode

- In the above figure, r_B represents Bulk Resistance of the Diode; r_p Resistance of the P Region; r_n represents Resistance of the N Region; r_j represents Resistance of the Junction
- The total resistance in a Diode is the sum of Bulk resistance and junction resistance

1.1. Forward Resistance

- For a forward biased PN junction, we can neglect the term I_0 in equation (4) and substitute I_F (forward current) instead of I

- Now the equation becomes,

$$g_j = \frac{I_F}{\eta V_T}$$

- Thus the forward resistance is given by

$$r_j = \frac{\eta V_T}{I_F} \quad (5)$$

- Substituting the values, the junction resistance for a forward biased junction is given as

$$r_j = 25 \text{ mV} / I_F (\text{mA}) \quad \text{for Ge} \quad (6)$$

$$r_j = 50 \text{ mV} / I_F (\text{mA}) \quad \text{for Si} \quad (7)$$

- Its value for forward-biased junction depends on the magnitude of forward dc current.
- From the above equation, it can be seen that it is a variable resistance
- The Forward Resistance is categorized into two;
 - (a) Static Resistance
 - (b) Dynamic Resistance

1.1.1. Static Resistance

- The resistance offered by a p-n junction diode when it is connected to a DC circuit is called static resistance.

1.1.2. Dynamic Resistance

- The dynamic resistance is the resistance offered by the p-n junction diode when AC voltage is applied.

1.2. Reverse Resistance

- Reverse resistance is the resistance offered by the p-n junction diode when it is reverse biased.
- When reverse biased voltage is applied to the p-n junction diode, the width of depletion region increases.
- This depletion region acts as a barrier to the electric current.
- Hence, a large amount of electric current is blocked by the depletion region.
- Thus, reverse biased diode offers large resistance to the electric current.
- The resistance offered by the reverse biased p-n junction diode is very large compared to the forward biased diode.
- The reverse resistance is in the range of megaohms (MΩ).
- Consider the equation

$$g_j = \frac{I_s e^{\frac{V}{\eta V_T}}}{\eta V_T} \quad (8)$$

- In the above equation when the reverse voltage V starts increasing from small values, the numerator term will be very small and the conductance will be very small.

- This means that the resistance r_j offered by the reverse biased PN junction is very high.

2. Junction Capacitance

- In a diode, Changes in voltage lead to changes in depletion width and charge.
- This leads to a capacitance that we can calculate from the charge-voltage dependence.
- There are two types of Capacitances present in a PN junction diode.
- They are
 - (a) Junction capacitance
 - (b) Diffusion capacitance.

2.1. Transition Capacitance or Space Charge Capacitance (C_T)

- Junction Capacitance comes from the depletion region.
- There is junction capacitance in both forward and reverse bias.
- When a P-N junction is reverse-biased, the depletion region acts like an insulator or as a dielectric material essential for making a capacitor.
- The P- and N-type regions on either side have low resistance and act as the plates.
- This combination forms a parallel plate capacitor.
- This junction capacitance is called transition or space charge capacitance (C_{pn} or C_T).
- It is calculated using the usual formula of Capacitance given by

$$C = \frac{\epsilon A}{d} \quad (9)$$

- Its typical value is 40 pF
- Since thickness of depletion (or transition) layer depends on the amount of reverse bias, capacitance C_T can be controlled with the help of applied bias.
- This property of variable capacitance possessed by a reverse biased P-N junction is used in the construction of a device known as Varicap or Varactor Diode.
- The voltage dependence of the Transition Capacitance is related using the following mathematical expression

$$C_T = \frac{K}{(V_K + V_R)^n} \quad (10)$$

- Where
 - V_K is the knee voltage of the Diode
 - V_R is the applied Reverse Voltage
 - K is the constant depending on the semiconductor material.
- $n = \frac{1}{2}$ for alloy junction
- $n = \frac{1}{3}$ for diffused junction
- The symbol and diagram of a Varicap is shown in Figure 2.

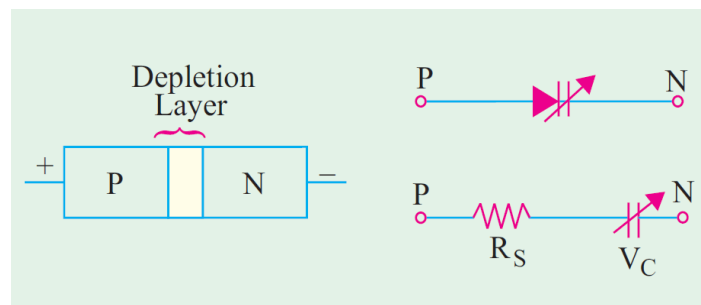


Figure 2. Transition Capacitance in a PN Junction Diode

2.2. Diffusion capacitance or Storage Capacitance (C_D)

- Diffusion capacitance only exists in forward bias.
- This is a nonlinear capacitor; and it is difficult to model.
- It is called diffusion capacitance because of the time delay in moving charges across the junction by diffusion process.
- It cannot be modelled using a parallel plate capacitor, because it varies directly with the magnitude of forward current.

2.2.1. How Diffusion Capacitance occurs in a PN junction

- Consider a forward-biased junction which is carrying a forward current I_F .
- Suppose the applied voltage is suddenly reversed, then I_F ceases suddenly but leaves a lot of majority charge carriers in the depletion region.
- These charge carriers must get out of the region.
- But due to the sudden reverse biasing, the depletion region becomes wider.
- Hence, it is seen that when a forward-biased P-N junction is suddenly reverse-biased, a reverse current flows which is large initially but gradually decreases to the level of saturation current I_0 .
- This effect can be likened to the discharging current of a capacitor and is, therefore, rightly represented by a capacitance called diffusion capacitance C_D .
- Since the number of charge carriers left in depletion layer is proportional to forward current, C_D is directly proportional to I_F .
- Typical value of Diffusion Capacitance is $0.02\mu\text{F}$ which is 5000 times C_T .
- The capacitance assumes great significance in the operation of devices which are required to switch rapidly from forward to reverse bias.
- If C_D is large, this switch-over cannot be rapid.
- It will delay both the switch-on and the switch-off.
- This effect of C_D is variously known as recovery time or carrier storage.

Reverse Breakdown in PN Junction Diodes

- If the reverse bias applied to a P-N junction is increased, a point is reached when the junction breaks down and reverse current rises sharply to a very high value
- At this point only the Resistor connected series to the junction can control or limit the flow of current
- This critical value of the voltage is known as "Breakdown Voltage" (V_{BK})
- Once breakdown has occurred, very little further increase in voltage is required to increase the current to relatively high values
- The junction itself offers almost zero resistance at this point.
- The breakdown voltage depends on the width of the depletion region which, in turn, depends on the doping level.
- There are two mechanisms by which Junction breakdown occurs in PN Junction diodes
- They are
 - Zener Breakdown
 - Avalanche Breakdown
- Figure 1 depicts the Current Voltage Characteristics of the two different breakdown mechanisms in PN Junction Diodes

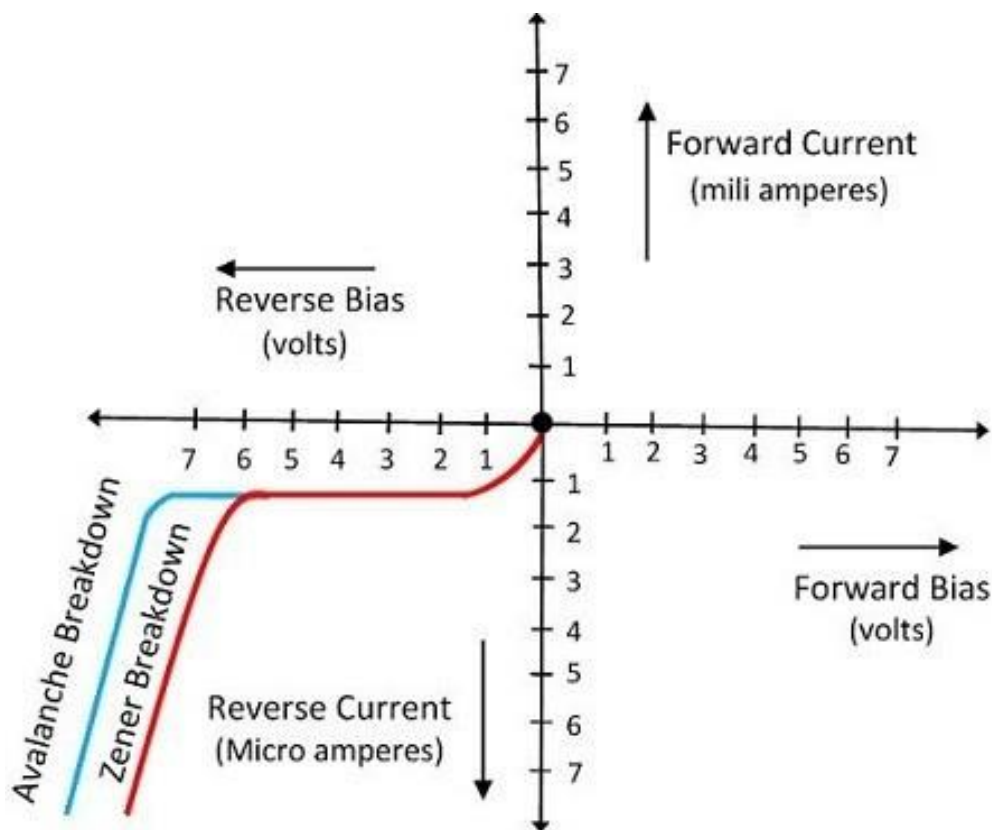


Figure 1. Breakdown Mechanisms in PN Junction diodes

1. Zener Breakdown

- This form of breakdown occurs in junctions which, is heavily doped and have narrow depletion layers.
- The breakdown voltage sets up a very strong electric field across this narrow layer.
- This field is strong enough to break or rupture the covalent bonds thereby generating electron-hole pairs.
- Even a small further increase in reverse voltage is capable of producing large number of current carriers.
- That is why the junction has very low resistance in the break-down region.
- The Mechanism of Zener Breakdown is shown in the below figure.

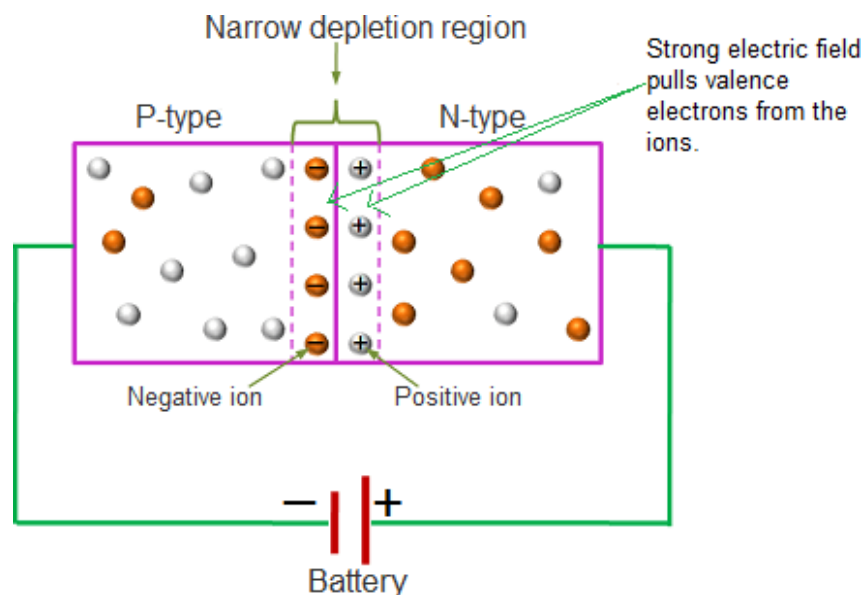


Figure 2. The Zener Breakdown Mechanism

2. Avalanche Breakdown

- This form of breakdown occurs in junctions which are lightly-doped and have wide depletion layers where the electric field is not strong enough to produce Zener breakdown.
- Here, the minority carriers (accelerated by this field) collide with the semiconductor atoms in the depletion region.
- Upon collision with valence electrons, covalent bonds are broken and electron-hole pairs are generated.
- These newly-generated charge carriers are also accelerated by the electric field resulting in more collisions and hence further production of charge carriers.
- This leads to an avalanche (or flood) of charge carriers and, consequently, to a very

low reverse resistance.

- The Avalanche Breakdown is shown in the below figure.

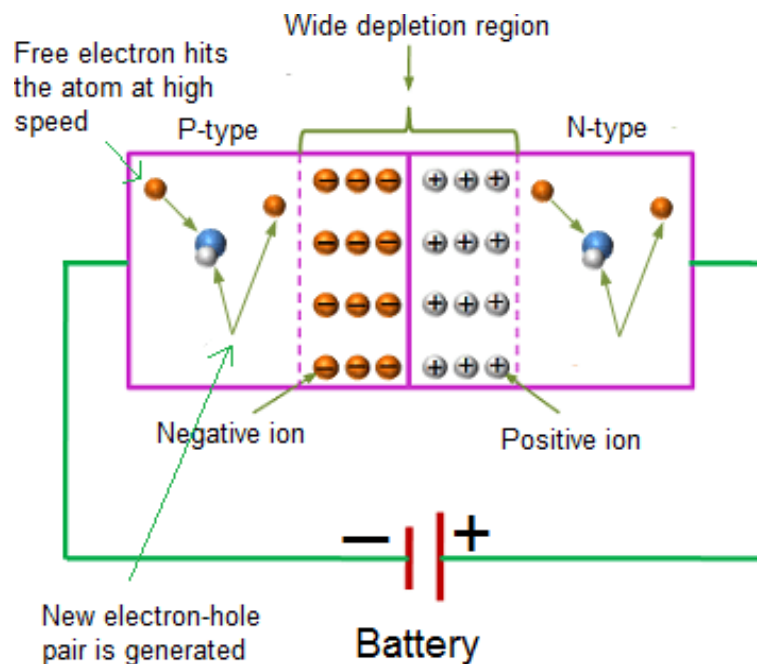


Figure3.TheAvalancheBreakdownMechanism

3. DifferencebetweenZenerBreakdownandAvalancheBreakdown

ZenerBreakdown	AvalancheBreakdown
Zener breakdown occurs because of the high electric field	Avalanche breakdown occurs because of the collision of the electrons
DepletionRegionisThin	DepletionRegionisThick
Junctionwillnotbedestroyed	Junctionwillbedestroyed
Strong Electric Field at a narrow depletion region need to be applied for this effect	Even a Weak Electric Field is enough for this effect
Heavydopingisrequired	Lightdopingisenough
Voltageremainsconstantafterbreakdown	Voltagevariesafterbreakdown
CovalentBondsarebrokenintheatoms	Atomsareionizedbytheelectrons
Occursatlowreversevoltatage	Occursathighreversevoltage

4. Applications of Reverse Breakdown Effect

- The diodes which use the Reverse Breakdown Mechanism are Zener Diode and Avalanche Diode
- The applications of such diodes in Electronics are given below

4.1. Zener Diode

- Voltage stabilizer or regulator
- Switching operations
- Peak clippers
- Surge suppressors

4.2. Avalanche Diode

- Voltage Reference
- RF noise generator
- Microwave frequency generator